

Wave guides as an analogous system of 1D optical lattice for anyons

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 (Dated: October 8, 2026)

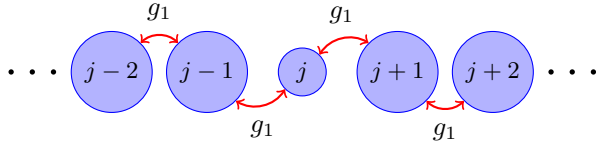
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Keywords: anyons, quantum matter, ultracold atoms

I. INTRODUCTION

Los anyones son ...

II. MAIN IDEA



Para las guías de onda:

- Una guía de onda posee un índice de refracción en el otro núcleo, rodeado por otro de menor índice $n_1 < n_2$
- Para una única guía de onda, los modos admitidos están dados por sus características físicas.
- Un arreglo de guías de onda, el acoplamiento permitirá la transferencia entre los elementos del arreglo.

From an infinite waveguides array with homogeneous optical single mode waveguides and equally spaced along z-axis. This is described by the following differential equation [1]

$$i \frac{dE_j(z)}{dz} = g_1(E_{j-1} + E_{j+1}), \quad j = -\infty, \dots, \infty, \quad (1)$$

From this

$$i \frac{d|\psi(z)\rangle}{dz} = \hat{H}|\psi(z)\rangle, \quad \hat{H} = g_1 (\hat{V}_\infty + \hat{V}_\infty^\dagger), \quad (2)$$

where the operators $\hat{V}_\infty$ and $\hat{V}_\infty^\dagger$ are the infinite counterparts of the London-Susskind-Glogower operators [2, 3], defined as

$$\hat{V}_\infty = \sum_{n=-\infty}^{\infty} |n\rangle\langle n+1|, \quad \hat{V}_\infty^\dagger = \sum_{n=-\infty}^{\infty} |n+1\rangle\langle n|, \quad (3)$$

For an analytical solution we need to satisfy in our Hilbert space the following conditions for the LSG operator

$$[\hat{V}_\infty, \hat{V}_\infty^\dagger] = 0, \quad \hat{V}_\infty \hat{V}_\infty^\dagger = \hat{I},$$

In this framework the initial state is a photon in the n site of an infinite array of waveguides $|\psi(0)\rangle = |n_0\rangle$;

$$|\psi_{n_0}(z)\rangle = \exp \left[-g_1 z \left(i \hat{V}_\infty^\dagger - \frac{1}{i \hat{V}_\infty^\dagger} \right) \right] |n_0\rangle.$$

Donde podemos identificar la función generadora de las funciones Bessel, para obtener

$$|\psi_{n_0}(z)\rangle = \sum_{n=-\infty}^{\infty} J_n(-2g_1 z) i^n \hat{V}_\infty^{\dagger n} |n_0\rangle.$$

para este modelo podemos hacer una analogía con un modelo de bosones en una red óptica partiendo del operador $\hat{V}_\infty$ y la evolución será escrita en función del tiempo:

$$\hat{V}_\infty = \sum_{j=-\infty}^{\infty} \hat{b}_j^\dagger \hat{b}_{j+1}, \quad \hat{V}_\infty^\dagger = \sum_{j=-\infty}^{\infty} \hat{b}_{j+1}^\dagger \hat{b}_j, \quad (4)$$

Rewriting the Hamiltonian as

$$\hat{H} = -J \sum_{j=-\infty}^{\infty} \left(\hat{b}_j^\dagger \hat{b}_{j+1} + \hat{b}_{j+1}^\dagger \hat{b}_j \right) \quad (5)$$

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The operator $\hat{V}_\infty^{\dagger n}$ is equivalent to create a particle in n site in the framework of optical lattices for a subspace of one particle per site:

$$\hat{V}_\infty^{\dagger n}|1_0\rangle = \hat{b}_n^\dagger|0\rangle, \quad (6)$$

Obtaining the dynamics for a particle in 0-site

$$|\psi_0(t)\rangle = \sum_{n=-\infty}^{\infty} J_n(-2Jt) i^n \hat{b}_n^\dagger |0\rangle. \quad (7)$$

We could obtain a rich dynamics for n -particles system working with a Heisenberg representation for one body operators $\hat{b}^\dagger$ applied in the sites, obtaining in general the expression:

$$\langle \hat{n}_j \rangle = \sum_{m=-\infty}^{\infty} |J_{m-j}(-2Jt)|^2 \langle \hat{n}_m(0) \rangle \quad (8)$$

$$\langle \mathbf{J}_j \rangle = iJ \langle \hat{b}_{j+1}^\dagger \hat{b}_j - \hat{b}_j^\dagger \hat{b}_{j+1} \rangle \quad (9)$$

$$\langle \mathbf{T}_j \rangle = \sum_{m=-\infty}^{\infty} |J_{m-j}(-2Jt)|^2 \langle \hat{n}_m(0) \rangle \quad (10)$$

also:

$$\mathbf{J} = iJ \sum_{i=1}^M \left(e^{-i\theta \hat{n}_i} \hat{b}_{i+1}^\dagger \hat{b}_i - \hat{b}_i^\dagger \hat{b}_{i+1} e^{i\theta \hat{n}_i} \right) \quad (11)$$

$$\mathbf{T} = J \sum_{i=1}^M \left(e^{-i\theta \hat{n}_i} \hat{b}_{i+1}^\dagger \hat{b}_i + \hat{b}_i^\dagger \hat{b}_{i+1} e^{i\theta \hat{n}_i} \right) \quad (12)$$

$$S(q) = \sum_{l=0}^M |J_l(-2Jt)|^2 - \sum_{l,m=0}^M e^{iq(l-m)} |J_l(-2Jt)|^2 |J_m(-2Jt)|^2$$

III. CONCLUSION

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Appendix: Appendix

In this paper we suggest to calculate the operator from active representations due the feasibility of operations

1. Passive representation of $\hat{n}_j$ operator

For this framework we need to specify the initial state, first, for 2 particles in the sites j, j' respectively, with $j \neq j'$, the state as function of time is

$$|\psi(t)\rangle = e^{-i\mathcal{H}t} |\psi(0)\rangle = e^{-i\mathcal{H}t} \hat{b}_j^\dagger \hat{b}_{j'}^\dagger |0\rangle, \quad (A.1)$$

as the temporal propagator is unitary we use the identity to get the following expressions

$$\begin{aligned} |\psi(t)\rangle &= e^{-i\mathcal{H}t} \hat{b}_j^\dagger \mathcal{I} \hat{b}_{j'}^\dagger \mathcal{I} |0\rangle \\ &= e^{-i\mathcal{H}t} \hat{b}_j^\dagger e^{i\mathcal{H}t} e^{-i\mathcal{H}t} \hat{b}_{j'}^\dagger e^{i\mathcal{H}t} e^{-i\mathcal{H}t} |0\rangle \\ &= e^{-i\mathcal{H}t} \hat{b}_j^\dagger e^{i\mathcal{H}t} e^{-i\mathcal{H}t} \hat{b}_{j'}^\dagger e^{i\mathcal{H}t} |0\rangle, \end{aligned} \quad (A.2)$$

as the bosons obey time invariance $\hat{U}^\dagger \hat{b}^\dagger \hat{U} = \hat{U} \hat{b}^\dagger \hat{U}^\dagger$

$$|\psi(t)\rangle = \hat{b}_j^\dagger \hat{b}_{j'}^\dagger |0\rangle, \quad (A.3)$$

we derive the Heisenberg equation of motion $\hat{b}^\dagger(t)$ from

$$i \frac{d}{dt} \hat{b}_j^\dagger = [\hat{b}_j^\dagger, \hat{H}], \quad (A.4)$$

doing the commutator with (5) obtain:

$$i \frac{d}{dt} \hat{b}_j^\dagger(t) = J \left(\hat{b}_{j-1}^\dagger + \hat{b}_{j+1}^\dagger \right), \quad (A.5)$$

consistent with the phenomenology of (1), to solve this differential equation we can use the Fourier transform $\hat{b}_j^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{-ikj} \hat{b}_k^\dagger$ and return to the original space with an inverse Fourier transform, so in momentum space the differential equation decouples:

$$i \frac{d}{dt} \hat{b}_k^\dagger(t) = 2J \cos(k) \hat{b}_k^\dagger(t), \quad (A.6)$$

Solving

$$\hat{b}_k^\dagger(t) = e^{-i2Jt \cos(k)} \hat{b}_k^\dagger(0), \quad (\text{A.7})$$

and taking the continuum limit to transform back to time space using the inverse Fourier transform

$$\hat{b}_j^\dagger(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ikj} \left(e^{-i2Jt \cos(k)} \hat{b}_k^\dagger(0) \right) dk, \quad (\text{A.8})$$

substituting $\hat{b}_k^\dagger(0) = \sum_m e^{ikm} \hat{b}_m^\dagger(0)$

$$\hat{b}_j^\dagger(t) = \sum_m \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-i2Jt \cos(k)} e^{ik(m-j)} \hat{b}_m^\dagger(0) dk \right), \quad (\text{A.9})$$

using the Jacobi Anger expansion:

$$e^{iz \cos(k)} = \sum_{n=-\infty}^{\infty} i^n J_n(z) e^{ink}. \quad (\text{A.10})$$

Substituting with $z = -2Jt$, we obtain

$$\hat{b}_j^\dagger(t) = \sum_{m,n} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} i^n J_n(-2Jt) e^{ik(m+n-j)} \hat{b}_m^\dagger(0) dk \right), \quad (\text{A.11})$$

due orthogonality of complex exponentials functions, the integral will be zero unless $n + m - j = 0$, so for the values $n = j - m$ and using properties of Bessel

function, we get obtain the the expression:

$$\hat{b}_j^\dagger(t) = \sum_{m=-\infty}^{\infty} i^{m-j} J_{m-j}(-2Jt) \hat{b}_m^\dagger(0) \quad (\text{A.12})$$

2. Active representation of $\hat{n}_j$ operator

We obtain for a $|\psi\rangle$ as steady state and $\hat{b}^\dagger$ as dynamic operator, write as:

$$\hat{b}_j^\dagger(t) = \sum_{m=-\infty}^{\infty} i^{m-j} J_{m-j}(-2Jt) \hat{b}_m^\dagger(0), \quad (\text{A.13})$$

so the expected value of $\hat{n}_j$ is:

$$\begin{aligned} \langle \hat{n}_j(t) \rangle &= \langle \psi(0) | \hat{b}_j^\dagger(t) \hat{b}_j(t) | \psi(0) \rangle, \\ &= \langle \psi(0) | \sum_{m,m'=-\infty}^{\infty} i^{m-m'} J_{m-j}(-2Jt) J_{m'-j}(-2Jt) \\ &\quad \hat{b}_m^\dagger \hat{b}_{m'} | \psi(0) \rangle, \end{aligned} \quad (\text{A.14})$$

In general we get a 1-d chain, if we annihilate at m' site and create at m we will obtain an orthogonal state, so for the different values of $m \neq m'$ we will obtain zero, so for an arbitrary initial state we get:

$$\langle \hat{n}_j(t) \rangle = \sum_{m=-\infty}^{\infty} |J_{m-j}(-2Jt)|^2 \langle \hat{n}_m(0) \rangle \quad (\text{A.15})$$